

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	Ragul.m	Reg. No.:	192524066
Section / Batch:	Slot A	Date:	24/07/2026

Code of Conduct

I RAGUL.M (192524066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: RAGUL.M

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

SIMATS Online Programming Lab
C++ PROGRAMMING

RAGUL M
102522000000

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Following Algorithm 5 marks

Q6 Binary Search 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

10/10 answered

Q1 Binary Search

Identify the error in the following snippet.

```

int binarySearch(int arr[], int N, int key) {
    int low = 0, high = N - 1;
    while (low < high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid;
        else
            high = mid;
        return -1;
    }
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

#include <iostream>
using namespace std;

int binarySearch(int arr[], int N, int key) {
    int low = 0, high = N - 1;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid;
    }
    return -1;
}

int main() {
    int arr[] = {10, 20, 30, 40, 50, 60, 70, 80, 90};
    int N = sizeof(arr) / sizeof(arr[0]);
    int key = 50;
    int ans = binarySearch(arr, N, key);
    if (ans != -1)
        cout << "Found. Index: " << ans << endl;
    else
        cout << "Not Found." << endl;
    return 0;
}

```

Submitted

Test Input

Enter Input (Optional)

Output

```

Found. Index: 50
Input: Index: 5

```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Following Algorithm 5 marks

Q6 Binary Search 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

10/10 answered

Q2 Stack

Find the error in the stack deletion operation. Justify it.

```

int pop() {
    if (top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

#include <iostream>
using namespace std;

const int MAX = 5;
int stack[MAX];
int top = -1;
int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top--];
}

void push(int x) {
    if (top == MAX - 1) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = x;
}

int main() {
    printf("Test1: pop() = %d\n", pop());
    push(10);
    printf("Test2: pop() = %d\n", pop());
    printf("Test3: pop() = %d\n", pop());
    printf("Test4: pop() = %d\n", pop());
    return 0;
}

```

Submitted

Test Input

Enter Input (Optional)

Output

```

Underflow
Test1: pop() = -1
Test2: pop() = 10
Test3: pop() = 10
Underflow

```

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DS PROGRAMMING

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

Q1
Stack Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Returning Symbol
5 marks

Q6
Single linked list
5 marks

Q7
Stack
5 marks

Q8
Linked list
5 marks

Q9
Linked list
5 marks

Q10
Stack
5 marks

10/10 answered

Q1 Stack
5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define stack 5
int stack[5];
int top;
void init() {
    top = -1;
}
void push() {
    printf("top = %d\n", top);
}
int main() {
    return 0;
}
```

Test Input

```
stdin: input (optional)
```

Output

```
top = -1
```

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INDONESIA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

Q1
Stack Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Returning Symbol
5 marks

Q6
Single linked list
5 marks

Q7
Stack
5 marks

Q8
Linked list
5 marks

Q9
Linked list
5 marks

Q10
Stack
5 marks

10/10 answered

Q2 Stack
5 marks

Identify the errors in the following snippet.

```
Stack = 0;
push(stack) > 0;
printf("%d", push);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define Stack 10
int Stack[Stack];
int top = -1;
void push(int data) {
    if (top < Stack-1) {
        Stack[++top] = data;
    }
}
void pop() {
    if (top >= 0) {
        printf("%d\n", Stack[top]);
        top--;
    }
}
int main() {
    int data;
    while (1) {
        printf("Enter data: ");
        scanf("%d", &data);
        push(data);
        printf("Stack: ");
        for (int i = 0; i <= top; i++) {
            printf("%d ", Stack[i]);
        }
        printf("\n");
        printf("Enter -1 to stop: ");
        scanf("%d", &data);
        if (data == -1) break;
    }
    return 0;
}
```

Test Input

```
stdin: input (optional)
```

Output

```
Stack: 0 0 0 0 0 0 0 0 0 0
```

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if(ch == ')' && stack[top] == '(') {  
    (ch == ')' && stack[top] == '(') {  
        pop();  
    } else {  
        return 0;  
    }  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>  
#include <stdlib.h>  
int isBalanced(char* s) {  
    int n = (strlen(s));  
    char stack[n];  
    int top = -1;  
    for (int i = 0; i < n; i++) {  
        char ch = s[i];  
        if (ch == '(' || ch == '[' || ch == '{') {  
            stack[++top] = ch;  
        } else if (ch == ')' || ch == ']' || ch == '}') {  
            if (top < 0) return 0;  
            char open = stack[top--];  
            if ((ch == ')' && open != '(') ||  
                (ch == ']' && open != '[') ||  
                (ch == '}' && open != '{')) {  
                return 0;  
            }  
        }  
    }  
    return (top == -1);  
}  
int main() {  
    char s[100];  
    printf("Enter a string: ");  
    scanf("%s", s);  
    if (isBalanced(s)) {  
        printf("Balanced\n");  
    } else {  
        printf("Not Balanced\n");  
    }  
    return 0;  
}
```

Submitted

Test Input

stdin input (optional)

Output

```
Test 1: "()" -> Balanced  
Test 2: "(())" -> Not  
Balanced  
Test 3: "()" -> Balanced  
Test 4: "(((((())" -> Balanced
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head  
while (temp->next != NULL) {  
    print(temp->data);  
    temp = temp->next;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>  
#include <stdlib.h>  
typedef struct Node {  
    int data;  
    struct Node *next;  
} Node;  
Node* createNode(int data) {  
    Node* node = (Node*) malloc(sizeof(Node));  
    node->data = data;  
    node->next = NULL;  
    return node;  
}  
Node* insertAtEnd(Node* head, int data) {  
    Node* node = createNode(data);  
    if (head == NULL) {  
        head = node;  
        return head;  
    }  
    Node* temp = head;  
    while (temp->next != NULL) {  
        temp = temp->next;  
    }  
    temp->next = node;  
    return head;  
}  
Node* buildListFromArray(int arr[], int n) {  
    Node* head = NULL;  
    for (int i = 0; i < n; i++) {  
        head = insertAtEnd(head, arr[i]);  
    }  
    return head;  
}  
void display(Node* head) {  
    Node* temp = head;  
    while (temp != NULL) {  
        printf("%d ", temp->data);  
        temp = temp->next;  
    }  
}
```

Submitted

Test Input

stdin input (optional)

Output

10 20 30 40

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• 1125

5. Results and discussion

🔗 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Test4 Input

stdin input completed

```

Thema 1: PHP mit mysqli anschauen
Thema 2: PDO anschauen
Thema 3: PDO mit PDOStatement

```

 Plotting statistical analysis

2019年12月15日

10. **एक संकेत**

🐛 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Test input

Test Input

stdin input (optional)

How is it done?	1	2	3	4
How often is it used?	4	5	6	7

► Providing results of reviews

Submit

← Back

10. **एक संकेत**

🐛 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Test input

Test Input

stdin input (optional)

How is it done?	1	2	3	4
How often is it used?	4	5	6	7

► Providing results of reviews

Submit

